The Register Is a Clock - IG

U06 · Practice · 24 written + 6 pen-first = 30 marks · Name: _____

Fresh authored questions. Work on paper. Use exactly eight bits unless another width is stated. Show working when requested. All marks are human marked; no computer is required.

1. Ours: read the temperature register

(a) Write +26 in eight-bit binary. Show your working. **[2]**

(b) Write -69 in eight-bit two's complement. Show your working. **[2]**

(c) The register contains `1011 0110`. Read it as a two's-complement integer. Show your working. **[2]**

2. Ours: the signed limits

State the smallest and largest integers that eight-bit two's complement can represent. **[2]**

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3. Ours: add the two counts

```
  0101 0011
+ 0010 1110
```

Perform the binary addition. Show carry working and the eight-bit answer. **[2]**

4. Ours: a counter with eight cells

```
  1011 0010
+ 0110 1101
```

(a) Perform the binary addition, showing your working. Give the eight bits stored if only the lowest eight bits are kept. **[2]**

(b) Explain why the true unsigned sum cannot fit in this register. Refer to the value and the available width. **[2]**

5. Ours: the label changes the meaning

Read `1000 0001` as (i) unsigned binary and (ii) two's complement. One mark per value. **[2]**

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6. Ours: zero and the final tick

In eight-bit two's complement, give (i) the bit pattern for zero and (ii) the denary value represented by `1111 1111`. **[2]**

7. Ours: the unfinished negative

A student writes +22 as `0001 0110`, inverts the bits to `1110 1001`, and calls this the two's-complement encoding of -22.

Identify the missing step and give the corrected eight-bit encoding. **[2]**

8. Ours: count is not the last label

State (i) how many distinct patterns eight bits allow and (ii) the largest unsigned integer represented by eight bits. **[2]**

9. Ours: does this one fit?

An unsigned eight-bit counter stores the sum of 99 and 28. State whether unsigned overflow occurs and justify your decision using the sum and the unsigned range. **[2]**

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10. Ours: the signed reader - pen first

6 marks - design before code. Given `bits`, a valid eight-character string of 0s and 1s, calculate its two's-complement denary value. The first column has weight -128; the others have weights 64, 32, 16, 8, 4, 2, 1. Store the answer in the integer `total`. Use selection, a loop and arithmetic; do not use `int(bits, 2)`, `bin()` or formatting conversions.
Optional after writing: copy your Q10(b) statements into the supplied `binary_to_denary` wrapper in `challenge_U06_Converter.py`. Its header and return line are provided; writing a function is not assessed.

(a) Write numbered steps or draw a flowchart showing how you treat the first bit and accumulate the remaining column values. **[2]**

(b) Write Python statements that calculate total from bits. You do not need to write input, output, a function header or a return statement. **[3]**

(c) State the total your program should produce for `1110 0011`. **[1]**

Above and beyond - outside the 30 marks. Use four-bit two's complement. Encode -8, invert and add one, keeping only four bits. Explain why the result cannot be +8. Now write -8 in eight bits and negate it again. What did the extra room change?
Optional code extension: complete the supplied reverse-converter body too. The file checks each direction against known values; passing the checks is not a written-method mark.

Additional space - label the question number.